

We are solving interencdec, an easy cryptography challenge where

I first download the file and run the 'file' command on it to get information on it:

```
(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$ file enc_flag
enc_flag: ASCII text

(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$
```

It tells me it is an ascii file, meaning I can read it in plaintext. I run the 'cat' command on it and got this:

```
(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$ cat enc_flag
YidkM0JxZGtwQlRYdHFhR3g2YUhsZmF6TnFLVGwzWVR0clh6ZzVNR3N5TXpjNWZRPT0nCg==

(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$
```

It looks like a base64 encoded string (based on the two equal signs at the end of it). I decoded it using the 'base64' command and got this:

```
(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$ base64 -d enc_flag
b'd3BqdkpBTXtqaGx6aHlfazNqeTL3YTNrXzg5MGsyMzc5fQ='
```

This output also looks like a base64 encoded string (the part in between the single quotes), so I decoded it using the 'base64' command again and got this:

```
(hackingmastermind@ctfdestroyingMACHINE)~/Downloads
$ echo "d3BqdkpBTXtqaGx6aHlfazNqeTL3YTNrXzg5MGsyMzc5fQ=" | base64 -d
wpjvJAM{jhlzhy_k3jy9wa3k_890k2379}
```

This string has the same format as the flag I am looking for. However, I had to figure out how to decode it. I didn't believe it was base64 encoded because it didn't have a double equals sign at the end of the string. It could be a Caesar cipher. The first letter is 'w', which is 7 letters away from 'p', the first letter of the flag. The second letter is 'p', which is 7 letters away from 'i', the second letter of the flag.

At this point, I thought I figured out how the flag was encoded. It was encoded using a substitution cipher where each letter of the flag was shifted 7 letters over. I crafted the following Python script to decode the flag:

```
(hackingmastermind@ctfdestroyingMACHINE) - [~/Downloads]
$ cat substitution_7.py
encrypted_flag = 'wpjvJAM{jhlzhy_k3jy9wa3k_890k2379}'

flag = ''

for i in encrypted_flag:
    if i in 'abcdefghijklmnopqrstuvwxyz':
        tmp = ord(i)-7
        if tmp < 97:
            flag += chr(123-(97-tmp))
        else:
            flag += chr(tmp)
    elif i in 'ABCDEFGHIJKLMNOPQRSTUVWXYZ':
        tmp = ord(i)-7
        if tmp < 65:
            flag += chr(91-(65-tmp))
        else:
            flag += chr(tmp)
    else:
        flag += i

print(flag)
```

Once I ran the script, I retrieved the flag:

```
(hackingmastermind@ctfdestroyingMACHINE) - [~/Downloads]
$ python3 substitution_7.py
picoCTF{caesar_d3cr9pt3d_890d2379}
```

flag: picoCTF{caesar_d3cr9pt3d_890d2379}